AUCKLAND, NEW ZEALAND · 36°50′56″S 174°45′36″E

Phil Donovan

Data Architecture · Engineering · Science · AI

DIRECTOR — VIRTUS SOLUTIONS

PRODUCTS — EOLAS.FYI

I design data platforms — Snowflake, dbt, Iceberg, AWS — and the models that run on them, for councils, health agencies and insurers. I came to it from geography, after ten years of spatial analysis across public health, transport and urban policy.

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Experience

2013 — PRESENT

2025 — now

Principal Consultant — Data Architecture, Engineering & AI

FIRN · AUCKLAND

- Led the design of Auckland Council's Modern Data Platform, centralising data assets for scale and governance.
- Built dbt, Snowflake and Fivetran pipelines — including automated ingestion of LINZ land parcels via WFS changesets, keeping council-wide spatial analysis current.
- Introduced and rolled out AI-assisted development tooling across the engineering team, setting the standards and workflows for its use.

2025 — now

Associate

PRINCIPAL ECONOMICS · AUCKLAND

- Built the Health Vulnerability and Adaptation dashboard for the Ministry of Health — an R Shiny and Plumber application modelling climate-change health impacts and adaptation options across New Zealand.
- Applied Treasury CBAx cost-benefit methodology, including declining discount rates, to cost adaptation scenarios down to SA2 level.
- Engineering on the Cost Intelligence Platform.

2024 — 2025

R/SQL Developer, Engineer & Scientist

AA INSURANCE · AUCKLAND

- Developed the pricing emulator that carried the migration from the legacy Protect platform to Duck Creek.
- Advised on the architecture of their AWS analysis platform, and built custom SageMaker container images so analysts were productive the moment one deployed.
- Wrote `aair`, an R package joining Redshift and other sources to the pricing emulator for reproducible pricing simulation.

2023 — 2024

Senior Analyst, R Developer & SQL Engineer

TE WHATU ORA — HEALTH NEW ZEALAND

- Led geospatial advisory for the Queenstown *Cryptosporidium* outbreak response, mapping contamination risk to direct public health intervention.
- Built `nphsintel`, an R package over Snowflake and PostGIS for population health analysis — cut code redundancy by around 75% and trained 10+ analysts on it.
- Introduced Quarto-based automated reporting and Shiny dashboards for spatial health trends.

2022 — 2023

Principal GIS Analyst

MINISTRY OF HEALTH · WELLINGTON

- Automated geospatial workflows with the `mohanalytics` and `mohgeospatial` R packages, lifting team productivity by around 40%.
- Built and maintained the ArcGIS Online Portal in Python and R, surfacing interactive layers on health access disparities with ETL keeping them current.
- Wrote monitoring scripts tracking portal credit and cloud spend, cutting resource overruns by around 30%.

2018 — 2020

Spatial Data Scientist

GEOGRAPHIC BUSINESS SOLUTIONS · AUCKLAND

- Deployed and configured the ESRI Enterprise stack — ArcGIS Pro and Server, Survey123, Collector, Operations Dashboard — for client geospatial platforms.
- Built 3D air pollution models for Seoul using the ArcGIS JavaScript API.

2016 — 2018

Spatial & Economic Data Analyst

MRCAGNEY · AUCKLAND

- Delivered the National Policy Statement on Urban Development Capacity analysis — regression modelling of the effect of zoning regulation on land prices nationally.
- Ran the Whangārei residential and business development capacity assessment, building geometric algorithms to encode planning rules.
- Public transport modelling against the Auckland and Wellington transport models.

2013 — 2016

Spatial Data Analyst

SHORE, MASSEY UNIVERSITY · AUCKLAND

- Principal spatial analyst on children's urban health research — from cleaning travel-diary data through to accessibility analysis.
- Co-authored five peer-reviewed papers on children's mobility and the built environment.

Capabilities

TOOLING

LANGUAGES

R · Python · SQL

WAREHOUSES & ENGINES

Snowflake · Redshift · PostgreSQL · Oracle · SQL Server · DuckDB · Apache Arrow · Iceberg

PIPELINES

dbt · Meltano · Fivetran · Singer · PySpark

CLOUD & INFRASTRUCTURE

AWS · Docker · Linux · Git

MODELLING

Regression · Supervised & unsupervised learning · Spatial models · Cost-benefit (CBAx)

SPATIAL

PostGIS · ArcGIS · QGIS · FME · GRASS

APPS & REPORTING

R Shiny · Quarto · Power BI · ArcGIS Portal

Selected work

OPEN SOURCE & PROJECTS

BUILDING NOW

eolas

Official statistics for New Zealand, Australia and the OECD are scattered across dozens of portals, each with its own formats and quirks. eolas puts 700+ datasets behind a single typed API — clean columns, consistent schemas, no merged Excel headers.

I built the whole stack: Singer-spec taps and a Meltano orchestration layer landing data as Iceberg tables on S3 and Glue, exposed through a FastAPI service on AWS, with first-class Python and R clients, a CLI, and a Snowflake share for warehouse users.

PYTHON · R · SINGER · MELTANO · ICEBERG · FASTAPI · SNOWFLAKE · AWS

nzcensr

An R package that makes New Zealand census data straightforward to import, in both spatial and non-spatial form.

R · SPATIAL

NZ Economic Overview

An interactive Shiny dashboard over the vs-warehouse API, surfacing New Zealand economic indicators.

R · SHINY

NPS-UDC

National Policy Statement on Urban Development Capacity — quantifying how local government regulation moves land prices across New Zealand's major urban areas, by regression on zoning rules.

ECONOMICS · SPATIAL REGRESSION

NDAI-C

The Network Destination Accessibility Index for Children: a tool measuring how reachable everyday destinations are for kids across Auckland, Hamilton, Wellington and Christchurch.

ACCESSIBILITY · NETWORK ANALYSIS

nphsintel

An R package for Te Whatu Ora giving analysts one route into population health data across Oracle and Snowflake, with spatial analysis built in. Cut code redundancy by around 75%.

R · SNOWFLAKE · POSTGIS

Writing

ARCHIVE · 2023 — 2024

16 May 2024

Enhancing efficiency with the `nphsintel` R package

Why an in-house R package was worth building, and how it cut analysis code by more than 75%.

7 January 2024

Utilising S3 classes in R for vaccine data analysis

Using R's oldest object system to make messy immunisation data behave.

29 November 2023

Packaging your day-to-day data analysis

Writing an R package is not reserved for virtuosos — it's the cheapest way to stop repeating yourself.

16 July 2023

Demystifying Apache Arrow with R

What Arrow actually is, and why it matters once your data outgrows memory.

Education

2007 — 2013

2012 — 2013

MSc in Geography — *First Class Honours*

University of Auckland

2011

PgDip in Geography

University of Auckland

2007 — 2010

BSc in Geography and Geology

Victoria University of Wellington

Published research

CO-AUTHORED

01 **Kids in the City: Children's Use and Experiences of Urban Neighbourhoods in Auckland, New Zealand**

Carroll, Witten, Kearns & Donovan

02 **Associations between the neighbourhood built environment and out of school physical activity and active travel**

Oliver, Mavoa, Badland, Parker, Donovan, Kearns, Lin & Witten

03 **Children's Out-of-School Independently Mobile Trips, Active Travel, and Physical Activity**

Oliver, Parker, Witten, Mavoa, Badland, Donovan, Chaudhury & Kearns

04 **Assessing neighbourhood destination access for children: development of the NDAI-C audit tool**

Badland, Donovan, Mavoa, Oliver, Chaudhury & Witten

05 **Development of a systems model to visualise the complexity of children's independent mobility**
Badland, Kearns, Carroll, Oliver, Mavoa, Donovan, Parker, Chaudhury, Lin & Witten

Get in touch

AUCKLAND · NZST

Open to conversations about data platform work, spatial architecture, and anything where the two meet.

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